

# Jhonathan Prieto Rojas, Ph.D.

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Links: [Google Scholar](#) | [Personal website](#) | [KFUPM Website](#)

## ACADEMIC PROFILE

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Electrical engineering academic with more than ten years of international experience in research-led teaching and interdisciplinary research. Expertise spans semiconductor devices, micro/nanofabrication, flexible and stretchable electronics, wearable sensing, thermoelectric energy harvesting, and self-powered electronic systems. Track record includes 37 peer-reviewed journal publications, five U.S. patents, four PI-led funded projects exceeding USD 85,000, and international collaborations involving KAUST, UIUC, Penn State and Georgia Tech. Experienced in developing and coordinating undergraduate and postgraduate courses in circuits, electronics, digital systems, semiconductor devices, micro/nanoelectronics, and sensors, as well as supervising research students, leading curriculum development, and supporting accreditation and academic quality assurance.

**Publications:** 37 peer-reviewed journal papers

**U.S. patents:** 5

**Total citations:** ~1575 (Scopus), ~2100 (Google Scholar)

**h-index:** 20 (Scopus), 24 (Google Scholar)

## PROFESSIONAL & RESEARCH EXPERIENCE

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### Assistant Professor | Sep. 2015 – Present

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Lead interdisciplinary research in semiconductor devices, micro/nanofabrication, flexible and stretchable electronics, thermoelectric energy harvesting, and self-powered wireless systems.
- Secured more than USD 85,000 as Principal Investigator across four research projects and contributed to three institutional and international collaborative projects.
- Develop and deliver undergraduate and postgraduate courses in circuits, electronics, digital systems, semiconductor devices, and micro/nanofabrication using research-led, project-based, and active-learning approaches.
- Supervise undergraduate capstones and postgraduate research, including two PhD projects, MSc theses, award-winning student prototypes, and projects leading to peer-reviewed publications.
- Lead and contribute to curriculum development, multi-section course coordination, ABET accreditation, academic quality assurance, graduate admissions, and faculty recruitment.

### Research Scholar — Concurrent appointment | May 2021 – Present

Interdisciplinary Research Center for Advanced Materials, KFUPM, Dhahran, Saudi Arabia

- Principal investigator on funded research projects in thermoelectric energy harvesting for wearable applications, and dust mitigation technologies.

### Postdoctoral Fellow | Oct. 2014 – Aug. 2015

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Development of novel micro/nano-fabrication techniques for high-performance and cost-effective flexible electronics. (More-than-Moore applications)
- Development of Thermoelectric Generators (TEGs) by incorporation of nano-engineered materials and sustainable designs.

### Visiting Researcher | May 2015 – Jun. 2015

University of Illinois at Urbana-Champaign, USA

Co-advisor: Prof. John Rogers

- Adaptation of transfer printing technology to improve the manufacturability of high-performance flexible electronics and enable 3D integration using flexible Si (100) fabrics.

### Graduate Research Assistant | Sep. 2009 – Jun. 2014

Integrated Nanotechnology Lab, KAUST, Thuwal, Saudi Arabia

- Development of novel micro/nano-fabrication techniques for high-performance and cost-effective flexible electronics. (More-than-Moore applications)
- Development of energy harvester/storage technologies by incorporation of nano-engineered materials and sustainable designs.

**Visiting Researcher, KAUST Global Collaborative Research (GCR) Program | Jun. 2010 – Aug. 2010**

The Pennsylvania State University, State College, USA

Co-advisor: Prof. Bruce Logan

- Design and successful fabrication of a micro-sized CNT-based Microbial Fuel Cell (MFC) with collaboration from the Materials Research Institute and Environmental Engineering Department.

**Intern, “Product-to-Market” Department | Mar. 2008 – Jul. 2008**

Micronas GmbH, Munich, Germany

- Evaluated, diagnosed, and corrected functionality issues in the hardware and software of an MPEG-2 video converter board mounted on a Spartan3 FPGA.
- Delivered a fully functional board with technical documentation after a five-months internship.

**EDUCATION**

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**King Abdullah University of Science and Technology, Saudi Arabia | Jan. 2011 – Jun. 2014**

Ph.D. Electrical Engineering, Electrophysics track (GPA: 3.84)

Dissertation Title: “Towards Flexible Self-Powered Micro-scale Integrated Systems”

Academic advisor: Prof. Muhammad M. Hussain, KAUST.

External PhD committee member: Prof. John Rogers, UIUC, USA.

**King Abdullah University of Science and Technology, Saudi Arabia | Sep. 2009 – Dec. 2010**

M.Sc. Electrical Engineering, Solid State Devices track (GPA: 3.93)

Thesis Title: “Advanced Nanofabrication Process Development for Self-Powered SoC”

Academic advisor: Prof. Muhammad M. Hussain

**Technical University of Munich, Munich, Germany | Oct. 2007 – Mar. 2008**

Exchange Student, Electronics Engineering (Focus on VLSI design, hardware programming and Nanotechnology)

**National University of Colombia, Colombia | Jan. 2003 – Mar. 2009**

B.Sc. in Electronics Engineering (GPA: 4.3/5, Ranked Top 3 in Engineering College)

Thesis Title: “Study and Optimization of a Development Card for CPLD XC9572XL”

**TEACHING AND ACADEMIC EXPERIENCE**

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- Participated in the development of three academic programs, leading one of them.
- Developed and taught three specialized senior/graduate courses in micro/nanoelectronics, semiconductor-device manufacturing, and micro/nanotechnology and biosensors.
- Coordinated multiple sections of circuits, electronics, and digital-logic courses and laboratories.
- Use active and project-based learning, team problem-solving, interactive polling, laboratory demonstrations, and research-linked capstone projects; recipient of the KFUPM Excellence in Teaching Award, 2020–2021.

**Academic Programs Developed:**

- Academic concentration (CX) in IC Design and Fabrications (EE, Phys, COE) (*Main Coordinator*)
- Academic concentration (CX) in Bioelectronics and Sensors (EE, Bio, COE) (*Team member*)
- Academic concentration (CX) in Mechatronics (ME, EE, COE) (*Team member*)

\* Each CX program is a four-course interdisciplinary concentration for senior students.

**Courses Developed & Taught:**

- EE436 – Introduction to Micro & Nanoelectronics
- EE439 – Micro/Nanotechnology & Biosensors
- EE649 – Solid State Devices & Manufacturing (Graduate level)

**Courses & Laboratories Taught:**

- EE200 – Digital Logic Circuit Design (Multi-section Coordinator)
- EE201 – Electric Circuits I (Multi-section Coordinator)
- EE203 – Electronics I (Multi-section Coordinator)
- EE203L – Electronics I Lab (Multi-section Coordinator)
- EE204 – Fundamentals of Electrical Circuits (Non-EE Students)
- EE234 – Electronics & Microcontrollers (Non-EE Students) (Multi-section Coordinator)

- EE236 – Electronic Circuits (Non-EE Students)
- EE303 – Electronics II (Multi-section Coordinator)
- EE304L – Electronics II Lab (Multi-section Coordinator)

## RESEARCH GRANTS

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Projects as Principal Investigator (Total funding secured > USD 85,000):

1. *Interdisciplinary Research Center Grant (~USD 20,000) | Feb. 2023 – Dec. 2024*  
Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase II]
2. *Interdisciplinary Research Center Grant (~USD 13,500) | Jul. 2021 – Dec. 2022*  
Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase I]
3. *KFUPM Internal Funded Grant (~USD 39,000) | Apr. 2017 – Jun. 2018*  
Advanced Study of Microfabricated Stretchable Silicon-based Structures
4. *KFUPM Junior Faculty Research Grant (~USD 12,800) | Feb. 2016 – Oct. 2016*  
Optimization of Spiral-based Structures for Stretchable Electronics

Projects as Co-Investigator:

1. *Georgia Tech-KFUPM Collaboration Grant | May 2019 – Nov. 2021*  
Design and Implementation of a Wireless Geophone Network for Seismic Data Collection
2. *KACST Technology Innovation Center - EE2381 | Jan. 2018 – Dec. 2020*  
Solid-State Lighting: Optical-based Sensing for Advanced Health Monitoring System
3. *KAUST-KFUPM Collaboration Grant | Oct. 2016 – Oct. 2019*  
Biometric data-based crowd management and risk mitigation for high population event

## RESEARCH SUPERVISION

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Primary supervisor of two completed MSc theses and two PhD dissertations in progress; committee member for two completed MSc theses; supervisor of ten senior-design projects and more than twelve industrial internships.

### Undergraduate:

- Advisor for 10 undergraduate senior design projects related to dust mitigation technologies, electronic sensing systems for geological exploration and wearable applications.
- Advisor for over 12 undergraduate students in their Industrial Internships and Summer Training experiences.
- Academic advisor to approximately 20 undergraduate students per semester, guiding degree planning and academic performance.

### Graduate:

M.Sc. Thesis Supervisor:

- Title: Flow-Field-Based Routing Algorithms for Efficient PCB Design Automation. (Completed, 2026)
- Title: Optimization of Spiral-based Structures for Stretchable Electronics. (Completed, 2017)

M.Sc. Committee Member:

- Title: Dynamics of MDOF Metamaterial Resonators Attached to a Periodic Waveguide. (Completed, 2022)
- Title: Probabilistic Sampling Towards Tunable Autonomous Sensing. (Completed, 2022)

PhD Dissertation Supervisor:

- Title: Hybrid Dust Mitigation Techniques for Solar Panels in Arid and Semi-Arid Regions. (In progress, Proposal defended, Expected graduation: June 2027)
- Title: Development of Kirigami-inspired Thermoelectric Generators for Self-powered Sensing Applications. (In progress, Proposal defended, Expected graduation: Dec. 2026)

## SELECTED JOURNAL PUBLICATIONS (Total: 37, 15 as first author, 6 as corresponding author, 7 cover pages)

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1. S. S. Ali, S. Firdous, S. Dieng, **J. P. Rojas**, T. A. Saleh, "MXene-assisted Bi<sub>2</sub>Te<sub>3</sub> pellet-based thermoelectric generator," *RSC Adv.*, published online August 13 (2026). (Q2)
2. **J. P. Rojas**, "Design and evaluation of LET-based soft hinges for stress mitigation in flexible copper interconnects", *Flex. Print. Electron.* 10, 035016 (2025). (Q2)
3. A. A. Alduais, A. B. Javaid, M. A. Syed, **J. P. Rojas**, "Analysis of Electrodynamical Screens with Fractal Designs for Enhanced Performance", *Arab. J. Sci. Eng.* 50, 17575-17585 (2025). (Q2)

4. **J. P. Rojas**, R. Almazyad, A. AlHayyah, A. Alruhaiman, M. Almusharraf, S. Al-Dharrab, H. Attia, "Self-Powered End-to-End Wireless Sensor Network for Geophysical Explorations," *IEEE Syst. J.* 19, 107-118 (2025). (Q1)
5. H. S. Daraghma, D. B. Ferry, S. G. Rao, M. A. Hawwa, M. A. Gondal, **J. P. Rojas**, "Materials & Design Approaches for Enhanced Performance of Mechanically Compliant Thermoelectric Generators (TEGs): A Review", *Smart Mater. Struct.* 33, 103003 (2024). (Q2)
6. M. S. Kim, A. S. Almuslem, W. Babatain, R. R. Bahabry, U. K. Das, N. El-Atab, M. Ghoneim, A. M. Hussain, A. T. Kutbee, J. Nassar, N. Qaiser, **J. P. Rojas**, S. F. Shaikh, G. A. Torres Sevilla, M. M. Hussain, "Beyond Flexible: Unveiling the Next Era of Flexible Electronic Systems", *Adv. Mater.* 36, 2406424 (2024). (Q1) **{Top viewed Article 2025 in Adv. Mater.}**
7. **J.P. Rojas**, M.U. Rehman, A. Hussein, A. E'mar, Y. AlRaei, H. AlZaher, M. Mohandes, "Kirigami-Enabled Wearable Health and Crowd Monitoring System", *Arab. J. Sci. Eng.* 47, 3583-3595 (2022). (Q2)
8. M.U. Rehman, W. Babatain, S.F. Shaikh, D. Conchouso, N. Qaiser, M. M. Hussain, **J. P. Rojas**, "Stress concentration analysis and fabrication of silicon (100) based ultra-stretchable structures with parylene coating", *Extreme Mech. Lett.* 41, 101052 (2020). (Q1)
9. M.U. Rehman, **J.P. Rojas**, "Optimization of compound serpentine-spiral structure for ultra-stretchable electronics", *Extreme Mech. Lett.* 15, 44-50 (2017). (Q1)
10. **J.P. Rojas**, D. Conchouso, A. Arevalo, D. Singh, I. G. Foulds, M. M. Hussain, "Paper-based origami flexible and foldable thermoelectric nanogenerator", *Nano Energy* 31; 296-301 (2017). (Q1)
11. **J.P. Rojas**, D. Singh, D. Conchouso, A. Arevalo, I. G. Foulds, M. M. Hussain, "Stretchable helical architecture inorganic-organic hetero thermoelectric generator", *Nano Energy* 30; 691-699 (2016). (Q1)
12. **J. P. Rojas**, G. A. Torres Sevilla, S. B. Inayat, A. M. Hussain, M. T. Ghoneim, S. Ahmed, M. M. Hussain, "Transformational Silicon Electronics", *ACS Nano* 8 (2), 1468-1474 (2014). (Q1)
13. J. E. Minkš, **J. P. Rojas**, B. E. Logan, M. M. Hussain, "Vertically Grown Multiwalled Carbon Nanotube Anode and Nickel Silicide Integrated High Performance Microsized (1.25  $\mu$ L) Microbial Fuel Cell", *Nano Letters* 12 (2), 791-795 (2012). *§These authors contributed equally to this work.* (Q1)

#### **SELECTED CONFERENCES AND PROCEEDINGS** (Total: 35, 12 as first author, 1 Best Presentation Award)

1. **J. P. Rojas**, PV Dust Removal through Fractal-based Electrodynamic Screens. **KAUST Research Conference 2023: Sustainable Energy Materials and Technologies for a Low Carbon Future.** (Invited talk)(2023)
2. **J. P. Rojas**, A. Hussein, H. AlZaher and M. Mohandes, Stretchable Wearable Crowd Monitoring System, **2021 4th Inter. Symp. on Adv. Elect. and Comm. Tech. (ISAECT)** (Paper Presentation) (2021)
3. A. Alduais, A. Javaid, M. Syed, M. Alhrtomi, **J. P. Rojas**, Fractal Based EDS Design for Increased Solar Panel Dust Removal Efficiency, **2021 MRS Fall Meeting & Exhibit.** (Paper Presentation) (2021)
4. **J. P. Rojas**, Origami and Kirigami stretchable electronics. **IEEE NEMS 2020** (Invited talk) (2020)
5. **J. P. Rojas**, Organic-inorganic heterostructures for stretchable electronics. **Proceeding of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications XI** 10982, 1098215 (Invited – Session Chair) (2019)
6. **J. P. Rojas**, M. U. Rehman, M. A. Albettar, D. Conchouso, A. Arevalo, D. Singh, I. Foulds, M. M. Hussain. "Folding and stretching a thermoelectric generator"; **Proceeding of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications X** 10639, 106390E (Invited – Session Chair) (2018)

#### **US PATENTS**

1. Cylindrical-Shaped Nanotube Field Effect Transistor, M. M. Hussain, H. M. Fahad, C. E. Smith, **J. P. Rojas**. US Patent US9224813B2.
2. Apparatus, system, and method for on-chip thermoelectricity generation, M. M. Hussain, H. M. Fahad, **J. P. Rojas**. US Patent 9515245B2.
3. Integrated circuit manufacturing for low-profile and flexible devices, M. M. Hussain, **J. P. Rojas**. US Patent US9209083B2.
4. Method for Producing Mechanically Flexible Silicon Substrate, M. M. Hussain, **J. P. Rojas**. US Patent US9520293B2.
5. Flexible and foldable paper-substrate thermoelectric generator (TEG) **J. P. Rojas**, M. M. Hussain. US Patent 20170244019A1

## PROFESSIONAL SERVICE

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### **Member, Assessment, Evaluation and Accreditation Committee | Jan. 2024 – Present**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Support the preparation of the Self-Study Report (SSR) for ABET accreditation.
- Coordinate the development and review of course specifications in the department.
- Assess the quality of teaching in the department.
- Support the development of guidelines for effective examination practices in the department.

### **Coordinator, Electronics Group, Electrical Engineering Dept. | Sep. 2021 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Coordination of the development and revision of academic programs, new courses/laboratories, and the acquisition of equipment for academic purposes.
- Coordinate the evaluation process of faculty and graduate student applicants.

### **Member, Faculty Search Committee | Sep. 2023 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation and interviewing of faculty applications, as well as lecturer and graduate assistant applicants.

### **Member, Academic Committee | Sep. 2023 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Manage the modification, update, and restructuring of EE undergraduate and graduate programs.
- Supervise the introduction of new courses and programs.
- Assess the quality of all EE programs and propose enhancement plans.

### **Treasurer, IEEE Western Saudi Arabia EDS Chapter | Nov. 2019 – Aug. 2023**

- Maintaining financial accounts and overseeing all fundraising efforts of the chapter.

### **Member, Graduate Admission Committee | Sep. 2015 – Jun. 2023**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation of graduate applications and recommendations to the graduate program.
- Review and update graduate needs for the department and identify targets and directions.

### **Chair, IEEE student branch | Jan. 2011 – Jun. 2014**

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.
- Meeting and hosting academic visitors.
- Support of several technical and social activities in the electrical engineering department.

### **Vice-chair, IEEE student branch | Sep. 2009 – Dec. 2010**

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.

## EDITORIAL AND REVIEWING ACTIVITIES

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### **Associate Editor | Jun. 2020 – Present**

Editorial duties in Frontiers in Electronics (JIF 2.1)

### **Guest Editor | Oct. 2018 – Jan. 2019**

Journal of Sensors (JIF 2.1)

Co-edited a special issue in “Flexible and/or Stretchable Sensor Systems”.

**Journal Reviewer for:** Int. J. Mech. Sci. (JIF 9.4), Bioresour. Technol. (JIF 9), Compos. Struct. (JIF 7.1), Appl. Therm. Eng. (JIF 6.9), Adv. Mater. Technol. (JIF 6.2), Meas. (JIF 5.6), IEEE Electron Device Lett. (JIF 4.5), Rev. Adv. Mater. Sci. (JIF 3.9), Energy Technol. (JIF 3.6), J. Mech. Behav. Biomed. Mater. (JIF 3.5), IEEE Trans. Electron Devices (JIF 3.2), Flex. Print. Electron. (JIF 3.2), IEEE Sens. Lett. (JIF 2.4), IEEE J. Flex. Electron. (CiteScore 5.2).

## HONORS AND AWARDS

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- Outstanding Alumni Award at International Level, Electrical and Electronics Engineers Association (AIEEUN), 60th Anniversary, National University of Colombia, 2022.
- *Second Place*: “Fractal-based Electrodynamic Screen”, 50K Challenge Award, KFUPM, May 2022. (Team advisor)
- Excellence in Teaching Award, College of Applied Engineering, KFUPM, Period 2020-2021.
- *First Prize*: “In-ear Health Monitoring System”, Hope Hackathon, Digital Health Track, Saudi Arabia, 2020. (Team advisor)
- *Awarded Second Prize* for “Novel Wafer Thinning Processing Based on  $\text{XeF}_2$  Etching for Flexible Electronics”, on the KAUST 1<sup>st</sup> Graduate Research Symposium, PSE division, May 4–5, 2011.
- *Best presentation Award* for “Carbon Nanotube based Micro-Sized Microbial Fuel Cell”, at the 5th Graduates’ Seminar at King Fahd University of Petroleum and Minerals, May 8, 2011.
- *KAUST Academic Excellence Award* (Fall 2010–Spring 2011).
- *KAUST Graduate Fellowship and Provost Award* 2009. (In recognition to students with outstanding scholastic achievement and personal accomplishment).
- *German Academic Exchange Service (DAAD) - Young Engineers Scholarship* 2007. (Awarded for excellent academic performance and German language skills).
- Participation at *VIII Iberoamerican Olympiad of Chemistry*. Cuernavaca, Mexico. October 2–9, 2003.
- Honorary Mention, 2002 *Colombian Physics Olympiad*.
- Gold Medal (2002), Honorary Mention (2001), *Colombian Chemistry Olympiad*.

## TECHNICAL SKILLS AND MEMBERSHIPS

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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>       | Spanish (native), English (fluent), German (Intermediate), Arabic (Basic).                                                                                                                                  |
| <b>Fab Skills</b>      | Photolithography, Wet Etch, Reactive Ion Etching, Atomic Layer Deposition (ALD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Sputtering, Evaporation, Annealing, Scanning electron microscopy (SEM). |
| <b>IT Skills</b>       | COMSOL, Coventor, L-Edit, MATLAB, LaTeX, C++, Visual Basic, VHDL, LabVIEW, Assembler.                                                                                                                       |
| <b>Education tools</b> | Blackboard, Blackboard Ultra, Poll Everywhere, Mentimeter, Gradescope.                                                                                                                                      |
| <b>Memberships</b>     | Member, IEEE; Member, IEEE Electron Device Society.                                                                                                                                                         |